

TUTORIAL INITIAL SETUP GEOCATCH 3D

1. Download python **3.13.2** dari <https://www.python.org/downloads/windows/>
Versi yang lebih baru belum stabil terkait rasterio / GDAL
<https://www.python.org/ftp/python/3.13.2/python-3.13.2-amd64.exe>

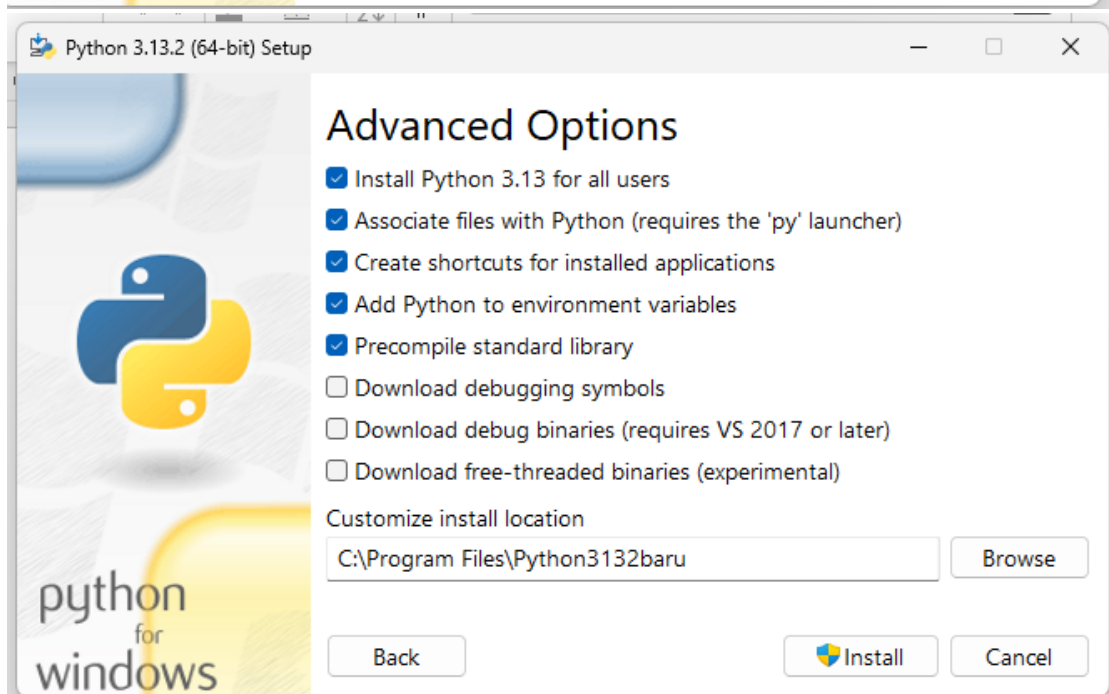
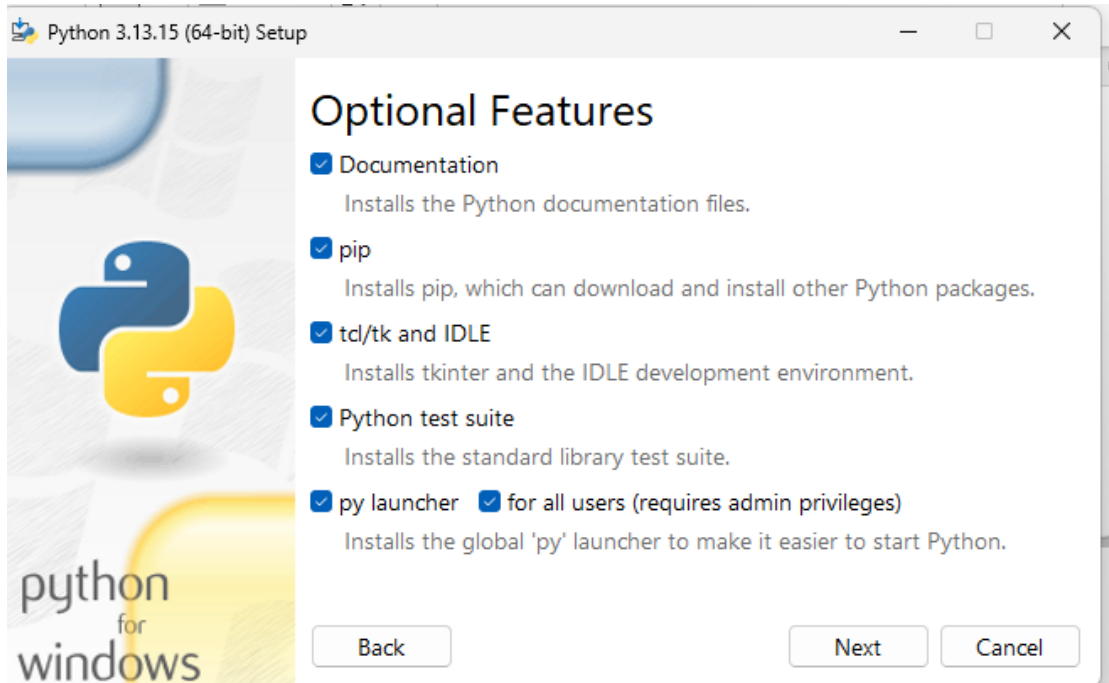
- Python **3.13.2** - Feb. 4, 2025

Note that Python **3.13.2 cannot be used on Windows 7 or earlier.**

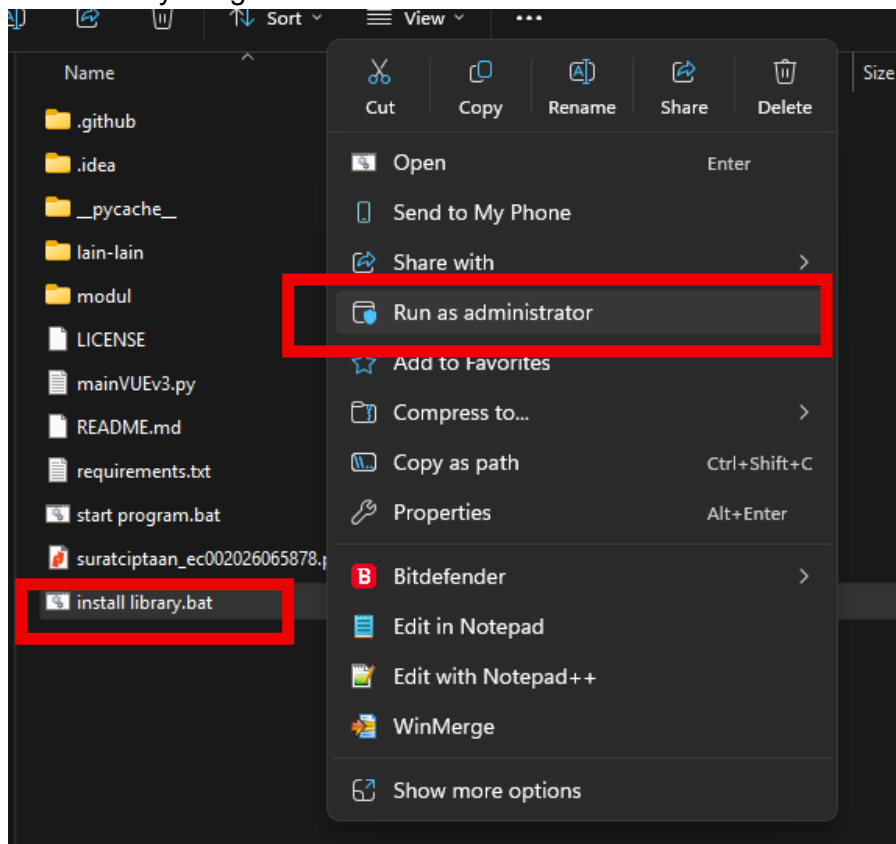
- Download Windows installer (64-bit)
- Download Windows installer (32-bit)
- Download Windows installer (ARM64)
- Download Windows embeddable package (64-bit)
- Download Windows embeddable package (32-bit)
- Download Windows embeddable package (ARM64)

2. Install pilih customize installation dan centang seperti gambar dibawah





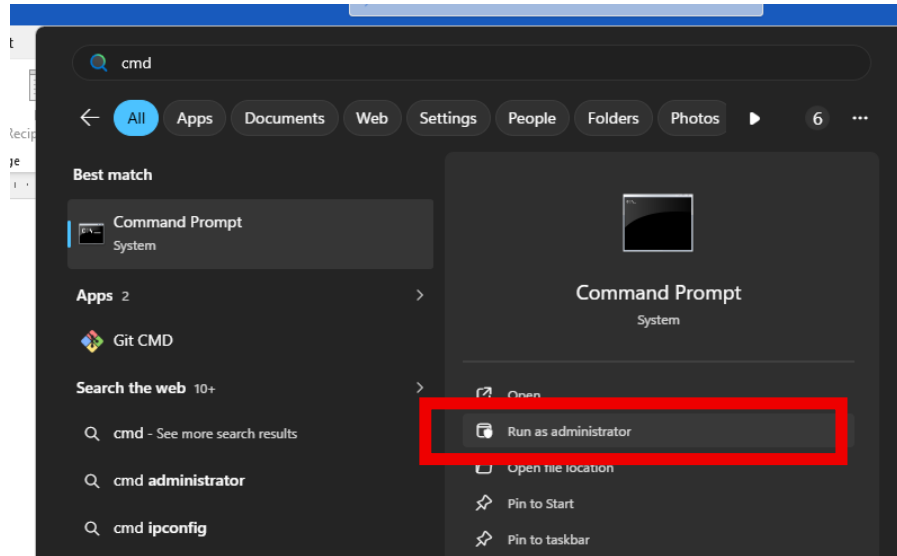
3. Install library dengan klik kanan run as administrator



```
C:\WINDOWS\System32\cmd.exe
2.19.0)
Requirement already satisfied: PyQt5-Qt5<5.16.0,>=5.15.2 in C:\Program Files\Python3132baru\Lib\site-packages (from PyQt5) (5.15.2)
Requirement already satisfied: colorama in C:\Program Files\Python3132baru\Lib\site-packages (from click!=8.2.*,>=4.0->rich) (0.4.6)
Requirement already satisfied: docstring-parser<4.0,>=0.15 in C:\Program Files\Python3132baru\Lib\site-packages (from cyclopts>=4.0.0->pyvista) (0.18.0)
Requirement already satisfied: rich-rst<3.0.0,>=1.3.1 in C:\Program Files\Python3132baru\Lib\site-packages (from cyclopts>=4.0.0->pyvista) (2.1.0)
Requirement already satisfied: rich>=13.6.0 in C:\Program Files\Python3132baru\Lib\site-packages (from cyclopts>=4.0.0->pyvista) (15.0.0)
Requirement already satisfied: pygments>=2.0.0 in C:\Program Files\Python3132baru\Lib\site-packages (from rich-rst<3.0.0,>=1.3.1->cyclopts>=4.0.0->pyvista) (2.20.0)
Requirement already satisfied: six>=1.5 in C:\Program Files\Python3132baru\Lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in C:\Program Files\Python3132baru\Lib\site-packages (from rich>=13.6.0->cyclopts>=4.0.0->pyvista) (4.2.0)
Requirement already satisfied: mdurl~=0.1 in C:\Program Files\Python3132baru\Lib\site-packages (from markdown-it-py>=2.2.0->rich>=13.6.0->cyclopts>=4.0.0->pyvista) (0.1.2)
Requirement already satisfied: platformdirs>=2.5.0 in C:\Program Files\Python3132baru\Lib\site-packages (from pooch->pyvista) (4.11.1)
Requirement already satisfied: requests>=2.19.0 in C:\Program Files\Python3132baru\Lib\site-packages (from pooch->pyvista) (2.34.2)
Requirement already satisfied: charset_normalizer<4,>=2 in C:\Program Files\Python3132baru\Lib\site-packages (from requests>=2.19.0->pooch->pyvista) (3.4.9)
Requirement already satisfied: urllib3<3,>=1.26 in C:\Program Files\Python3132baru\Lib\site-packages (from requests>=2.19.0->pooch->pyvista) (2.7.0)

D:\GeoCatch3D-main\GeoCatch3D-main>pause
Press any key to continue . . .
```

4. Jika langkah 3 berhasil loncat ke langkah 6.
Alternatif Buka cmd sebagai administrator



5. Jalankan perintah
`pip install pyvista trame trame-vuetify trame-vtk rasterio whitebox matplotlib PyOpenGL PyQt5`

Two screenshots of a Windows Command Prompt. The top screenshot shows the command prompt window with the command `C:\Users\agung>pip install pyvista trame trame-vuetify trame-vtk rasterio whitebox matplotlib PyOpenGL PyQt5`. The bottom screenshot shows the output of the command, listing the installed packages and their versions. The output includes a list of installed packages and their versions, followed by a list of packages that were successfully installed. The output is as follows:

```
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

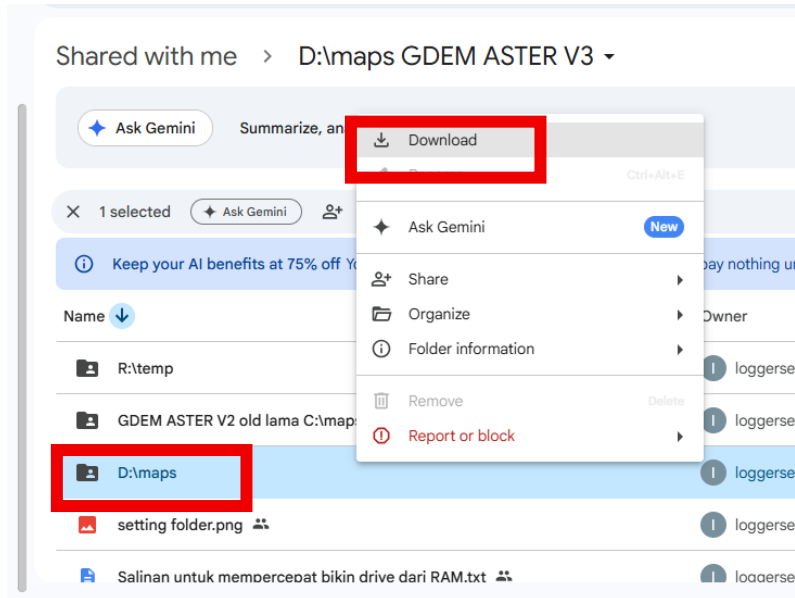
C:\Users\agung>pip install pyvista trame trame-vuetify trame-vtk rasterio whitebox matplotlib PyOpenGL PyQt5

Using cached aiosignal-1.4.0-py3-none-any.whl (7.5 kB)
Using cached charset_normalizer-3.4.9-cp313-cp313-win_amd64.whl (161 kB)
Using cached frozenlist-1.8.0-cp313-cp313-win_amd64.whl (43 kB)
Using cached idna-3.18-py3-none-any.whl (65 kB)
Using cached markdown_it_py-4.2.0-py3-none-any.whl (91 kB)
Using cached multidict-6.7.1-cp313-cp313-win_amd64.whl (45 kB)
Using cached propcache-0.5.2-cp313-cp313-win_amd64.whl (41 kB)
Using cached pygments-2.20.0-py3-none-any.whl (1.2 MB)
Using cached urllib3-2.7.0-py3-none-any.whl (131 kB)
Using cached yarl-1.24.5-cp313-cp313-win_amd64.whl (97 kB)
Using cached mdurl-0.1.2-py3-none-any.whl (10.0 kB)
Installing collected packages: PyQt5-Qt5, PyOpenGL, urllib3, typing-extensions, trame-common, six, scooby, pyyaml, PyQt5-sip, pyparsing, pygments, propcache, platformdirs, pillow, packaging, numpy, multidict, msgpack, more-itertools, mdurl, kiwisolver, idna, frozenlist, fonttools, docstring-parser, cyclopts, colorama, charset_normalizer, certifi, attrs, aiohappyeyeballs, yarl, trame-client, requests, python-dateutil, PyQt5, markdown-it-py, contourpy, click, aiosignal, affine, whitebox, trame-vuetify, trame-vtk, rich, rasterio, pooch, matplotlib, aiohttp, wslink, vtk, rich-rst, trame-server, cycl
opts, trame, pyvista
Successfully installed PyOpenGL-3.1.10 PyQt5-5.15.11 PyQt5-Qt5-5.15.2 PyQt5-sip-12.19.0 affine-3.0.0 aiohappyeyeballs-2.
7.1 aiohttp-3.14.3 aiosignal-1.4.0 attrs-26.1.0 certifi-2026.7.22 charset_normalizer-3.4.9 click-8.4.2 colorama-0.4.6 co
ntourpy-1.3.3 cyclopts-0.12.1 cyclopts-4.22.5 docstring-parser-0.18.0 fonttools-4.63.0 frozenlist-1.8.0 idna-3.18 kiwisolv
er-1.5.0 markdown-it-py-4.2.0 matplotlib-3.11.1 mdurl-0.1.2 more-itertools-11.1.0 msgpack-1.2.1 multidict-6.7.1 numpy-2.
5.1 packaging-26.3 pillow-12.3.0 platformdirs-4.11.1 pooch-1.9.0 propcache-0.5.2 pygments-2.20.0 pyparsing-3.3.2 python-
dateutil-2.9.0.post0 pyvista-0.48.4 pyyaml-6.0.3 rasterio-1.5.1 requests-2.34.2 rich-15.0.0 rich-rst-2.1.0 scooby-0.11.2
six-1.17.0 trame-3.13.2 trame-client-3.13.5 trame-common-1.2.7 trame-server-3.13.2 trame-vtk-2.11.15 trame-vuetify-3.2.
5 typing-extensions-4.16.0 urllib3-2.7.0 vtk-9.6.2 whitebox-2.3.6 wslink-2.5.7 yarl-1.24.5

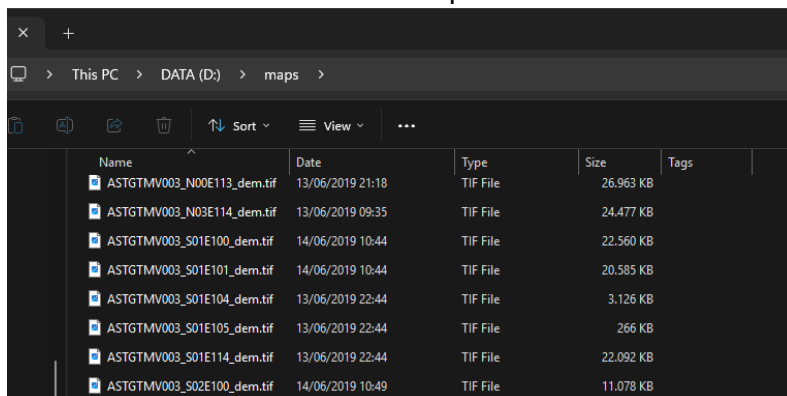
[notice] A new release of pip is available: 24.3.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Windows\System32>
```

6. Download data GDEM ASTER V3 wilayah Indonesia dari <https://drive.google.com/drive/folders/1I1kSE6V8pxy0Ox9FnMJopjOLXfiUxmKI>



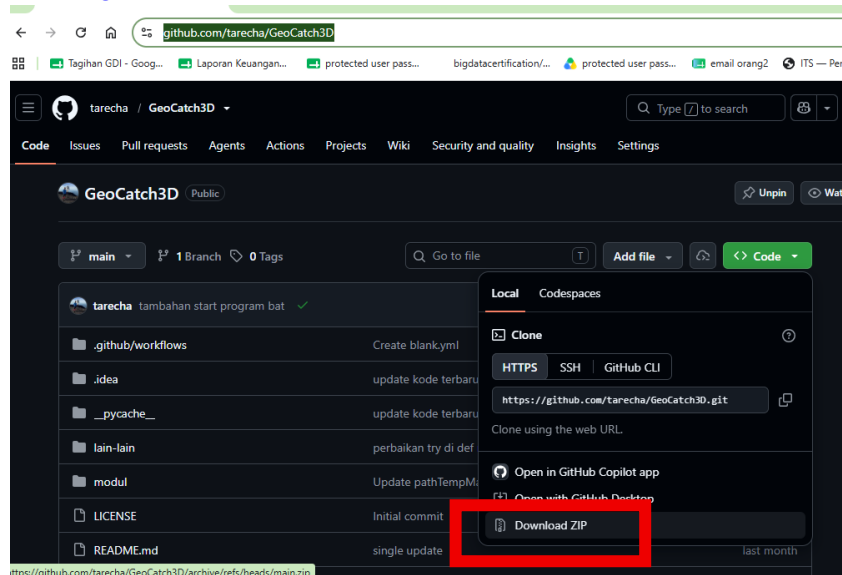
Kemudian ekstrak ke folder D:\maps



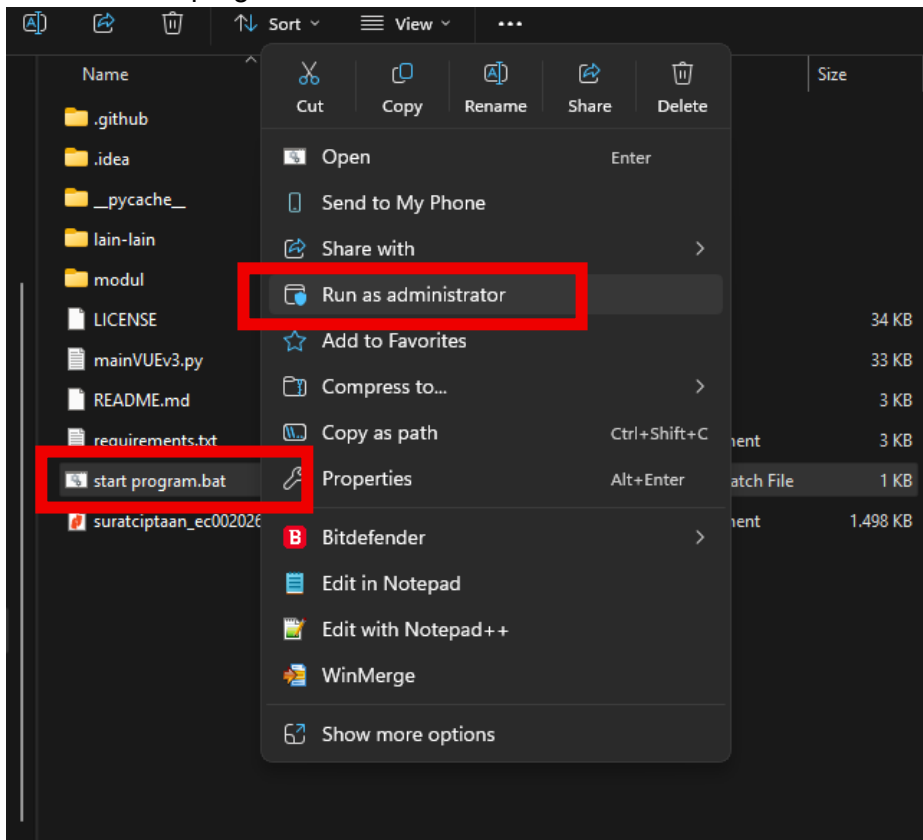
Catatan bisa tidak di download semua namun download file dengan koordinat sesuai lokasi yang diinginkan saja.

7. Download source code GeoCatch 3D, pada bagian code download ZIP kemudian ekstrak .zip nya

<https://github.com/tarecha/GeoCatch3D>



8. Jalankan start program.bat run as administrator



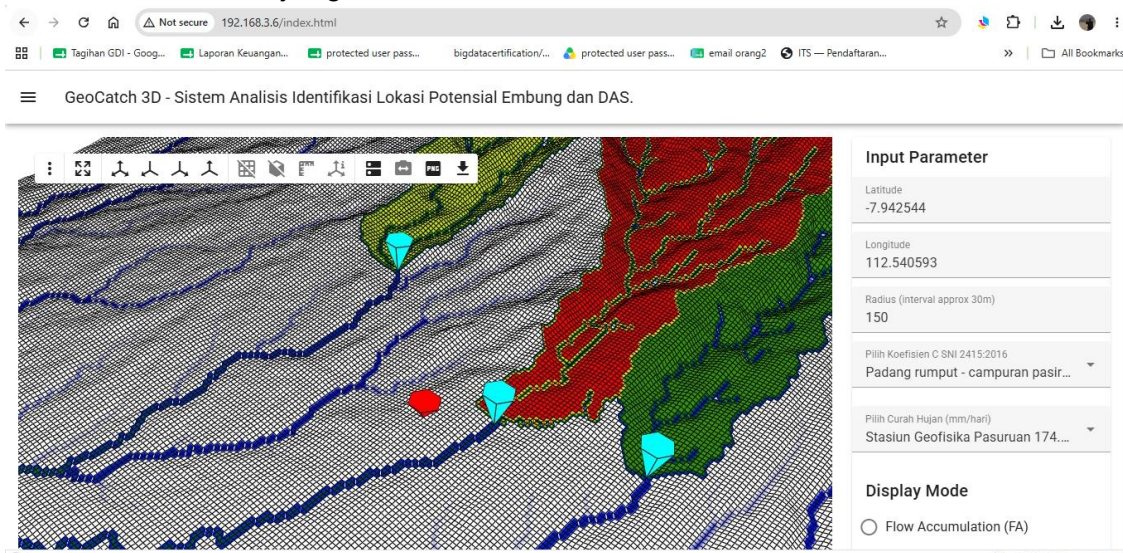
```
C:\WINDOWS\System32\cmd.exe
C:\Windows\System32>cd /d "D:\GeoCatch3D-main\GeoCatch3D-main\"

D:\GeoCatch3D-main\GeoCatch3D-main>python mainVUEv3.py --server
Downloading WhiteboxTools pre-compiled binary for first time use ...
Downloading WhiteboxTools binary from https://www.whiteboxgeo.com/WBT_Windows/WhiteboxTools_win_amd64.zip
Decompressing WhiteboxTools_win_amd64.zip ...
WhiteboxTools package directory: C:\Program Files\Python3132baru\Lib\site-packages\whitebox
Downloading testdata ...
Akses melalui web browser dari PC lain dengan link berikut :
-> CTRL + klik http://192.168.3.6

App running at:
- Local: http://0.0.0.0:80/
- Network: http://0.0.0.0:80/

Note that for multi-users you need to use and configure a launcher.
```

9. Akan muncul alamat yang bisa diakses melalui browser dan otomatis membuka browser



10. Optional untuk membuat drive template file dari ram untuk mempercepat bikin drive dari RAM

<https://sourceforge.net/projects/aim-toolkit/>

namun bila tidak ingin bikin drive dari RAM cukup rubah modul\config.py

pathTempMaps = r'R:\temp'

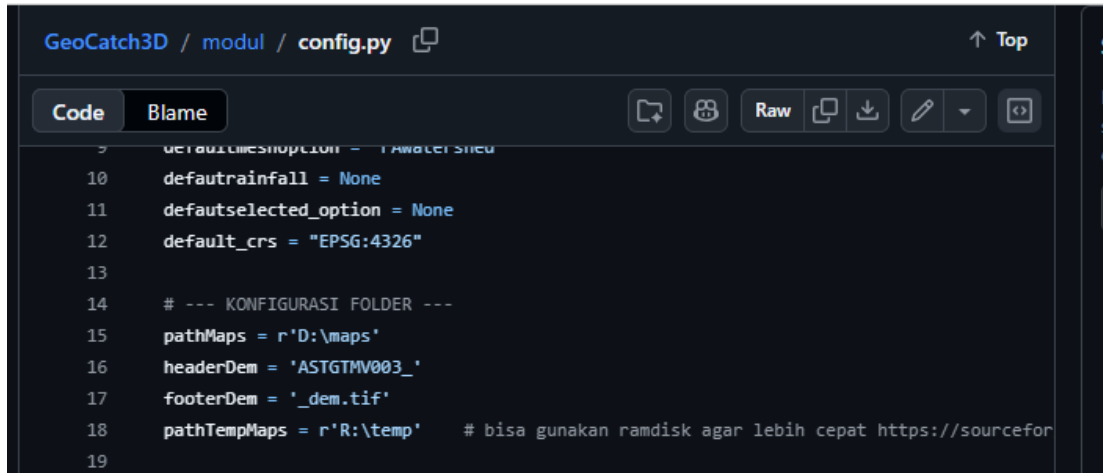
→ jika bikin ramdrive

ke

pathTempMaps = r'D:\maps\temp'

→ jika tidak bikin ramdrive

bisa drive apa saja disesuaikan. bikin drivenya dulu



```
GeoCatch3D / modul / config.py
9 defaultwateroption = 'water.shp'
10 defaultrainfall = None
11 defaultselected_option = None
12 default_crs = "EPSG:4326"
13
14 # --- KONFIGURASI FOLDER ---
15 pathMaps = r'D:\maps'
16 headerDem = 'ASTGMTV003_'
17 footerDem = '_dem.tif'
18 pathTempMaps = r'R:\temp' # bisa gunakan ramdisk agar lebih cepat https://sourcefor
19
```

Di Linux, Anda tidak memerlukan aplikasi pihak ketiga untuk membuat storage (penyimpanan) dari RAM, karena sistem secara bawaan sudah menyediakan fitur kernel bernama tmpfs. Metode ini jauh lebih cepat dibandingkan RAM disk tradisional (seperti ramfs) karena bisa diperbesar atau diperkecil secara dinamis dan ruang yang tidak terpakai bisa dikembalikan ke sistem. Untuk membuat storage RAM menggunakan tmpfs melalui terminal:

Buat folder sementara di direktori Anda (misal di folder Mount default mnt):

```
sudo mkdir -p /mnt/ramdisk
```

Pasang (Mount) partisi tmpfs dengan ukuran yang Anda inginkan (contoh: 2 GB):

```
sudo mount -t tmpfs -o size=2G none /mnt/ramdisk
```

Cek apakah sudah berhasil dengan perintah:

```
df -h /mnt/ramdisk
```

<https://www.scaler.com/topics/linux-ramdisk/>